

VERDEXIS

Product Specification Document

AI-Powered Multi-Asset Trading & Banking Platform

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1. Executive Summary

Verdexis is a next-generation fintech platform that unifies cryptocurrency trading, portfolio management, digital banking, and AI-powered financial analysis into a single, cohesive user experience. Designed for both retail and institutional investors, the platform combines real-time market data from multiple exchanges with intelligent portfolio insights to help users make informed financial decisions.

The platform addresses the fragmentation problem in today's fintech landscape, where investors typically need separate applications for trading (TradingView, Robinhood), banking (Fidelity, Chase), and research (CoinGecko, Bloomberg). Verdexis eliminates this friction by providing all essential financial services in one unified interface.

Key differentiators include: real-time crypto price data via CoinGecko API with fallback demo data; an AI-powered financial analyst that generates portfolio insights and market alerts; a functional trading simulator with localStorage-persisted portfolio data; a digital wallet supporting deposits, withdrawals, and transfers; and a dark-mode-first design with glassmorphism UI elements optimized for extended trading sessions.

2. Product Overview

2.1 Product Name & Identity

Product Name: Verdexis (pronounced "ver-DEX-is")

Brand Tagline: "Trade Smarter. Bank Seamlessly."

Visual Identity: The brand uses an emerald green (#0C8B44) as the primary color on a deep charcoal (#070C0E) background. The design language emphasizes glassmorphism cards with subtle transparency, clean geometric shapes, and a dark-mode-first aesthetic that reduces eye strain during extended trading sessions.

2.2 Product Purpose

Verdexis serves as an all-in-one financial command center. Its primary purposes are:

- Provide real-time access to cryptocurrency market data across 20+ top coins
- Enable portfolio tracking with P&L analytics and allocation breakdowns
- Simulate trading with market/limit orders and track trade history
- Manage a digital multi-currency wallet with deposit/withdraw/transfer capabilities
- Deliver AI-generated market insights, trend alerts, and portfolio recommendations

- Aggregate financial news from multiple sources in a single feed

2.3 Target Audience

Segment	Profile	Key Needs
Retail Crypto Investors	Individuals aged 25-45 holding \$5K-\$500K in crypto assets	Real-time prices, portfolio tracking, simple trading
Active Traders	Day/swing traders monitoring multiple assets daily	Advanced charts, order types, quick execution
Long-term Holders	Investors with diversified portfolios across multiple assets	P&L tracking, allocation analysis, news aggregation
Fintech Enthusiasts	Early adopters interested in AI-powered financial tools	AI insights, market predictions, novel features
Financial Advisors	Professionals managing client portfolios	Comprehensive dashboard, reporting, multi-asset view

3. Feature Specifications

3.1 Landing Page

The landing page serves as the primary marketing and entry point for new users. It is designed to convert visitors into registered users through clear value propositions and social proof.

Key Sections:

- Hero Section: Full-screen intro with animated background, platform headline, CTA buttons (Sign Up / Log In), and real-time market ticker
- Stats Bar: Animated counters showing platform metrics (users, volume, assets supported, uptime)
- Partner Logos: Frosted glass pill containers displaying integration partners (CoinGecko, Plaid, Stripe, Binance, Alpha Vantage, Finnhub)
- How It Works: 4-step visual guide (Connect Wallet, Fund Account, Trade Assets, AI Insights)
- Live Market Preview: Real-time price cards for top 6 cryptocurrencies with sparkline charts
- Testimonials: 4 customer testimonials with photos, ratings, and company names
- Pricing Plans: Three tiers (Starter \$0, Pro \$29/mo, Institutional \$99/mo) with feature comparison
- Security Section: Compliance badges (SOC 2 Type II, ISO 27001, PCI DSS, GDPR) with trust messaging

- FAQ Accordion: 6 common questions about security, fees, supported assets, and account types
- Footer: Logo, navigation links, social icons, newsletter signup, legal links

3.2 Authentication System

The authentication system provides secure account access with multiple login options and session management.

Features:

- Email/Password Registration: Full name, email, password with validation (min 8 characters)
- Email/Password Login: Credentials with "Remember Me" option
- Google OAuth: One-click login via Google account
- Session Persistence: Auth state stored in localStorage with automatic session recovery
- Protected Routes: Dashboard, AI Analyst, and Wallet pages require authentication
- Logout: Full session cleanup with page refresh
- Trust Indicators: SSL encryption badge, 2FA ready messaging

3.3 User Dashboard

The dashboard is the central hub for authenticated users, providing a comprehensive overview of their financial position and market conditions.

Components:

- Portfolio Value Card: Total net worth with 30-day SVG area chart, percentage change indicator
- Stats Row: 4 KPI cards — Total Value, 24h Change, Best Performer, Holdings Count
- Quick Actions: Deposit, Withdraw, Trade, and AI Insights shortcut buttons
- AI Insights Panel: Live AI-generated recommendations with confidence scores
- Portfolio Breakdown: Holdings list with real crypto logos, donut chart allocation, and allocation bars
- Recent Trades: Last 5 trades with buy/sell indicators, quantities, and prices
- Wallet Balances: Multi-currency wallet with coin logos and recent transaction activity
- Market Overview: 6 crypto cards with real logos, prices, 24h change, and SVG sparklines

3.4 Trading Interface

The trading page provides a professional-grade interface for executing trades and monitoring market conditions.

Features:

- Market List: 20+ cryptocurrencies with real logos, prices, 24h change, and market cap
- Search: Real-time filtering by name or symbol
- Watchlist: Star/unstar assets for quick access (persisted in localStorage)
- Order Form: Market/limit orders with buy/sell toggle, amount input, price estimation
- Balance Display: Available USD and selected crypto balance (gated behind auth)
- SVG Candlestick Chart: Real OHLC data rendered as proper candlestick bodies with wicks
- Time Range Selector: 1D, 1W, 1M, 3M, 1Y views
- Order Book: Simulated bid/ask spread with depth visualization
- Recent Trades: Personal trade history with timestamps
- Trade Execution: Functional buy/sell with portfolio updates and toast notifications

3.5 Wallet & Banking

The wallet module provides digital banking functionality for managing fiat and crypto balances.

Features:

- Multi-Currency Balances: USD, BTC, ETH, SOL with real-time values
- Deposit: Bank transfer simulation with amount input and confirmation
- Withdraw: Cash out to linked bank account with processing status
- Transfer: Send between internal wallets with fee estimation
- Transaction History: Chronological list with type icons, amounts, descriptions, and status
- Total Balance Card: Aggregate portfolio value across all currencies

3.6 AI Financial Analyst

The AI Analyst provides intelligent market analysis and portfolio recommendations powered by rule-based analytics engine.

Capabilities:

- Portfolio Insights: Analyzes holdings to generate personalized recommendations

- Market Alerts: Detects significant price movements and trend changes
- Technical Analysis: Identifies support/resistance levels and momentum signals
- Confidence Scoring: Each insight includes a 0-100% confidence indicator
- Chat Interface: Interactive Q&A for market and portfolio questions
- Real-time Data: Insights refresh based on live CoinGecko market data

3.7 Market News Feed

The news page aggregates financial news from multiple sources for comprehensive market coverage.

Features:

- Category Filters: Crypto, Stocks, Macro, Technology tabs
- Article Cards: Headline, source, summary, timestamp for each story
- Real-time Updates: Fetches from Finnhub API with fallback to curated demo content
- External Links: Direct links to full articles on source websites

4. Technical Architecture

4.1 Frontend Stack

Layer	Technology	Purpose
Framework	React 19 + TypeScript	UI component architecture with type safety
Build Tool	Vite 7.x	Fast development and optimized production builds
Styling	Tailwind CSS 3.4 + shadcn/ui	Utility-first CSS with 40+ pre-built components
Routing	React Router v6	Client-side navigation with protected routes
Charts	SVG (custom)	Candlestick charts, area charts, sparklines, donut charts
Animations	CSS transitions	Smooth UI transitions without heavy animation libraries
Notifications	Sonner	Toast notifications for trade confirmations and errors

Layer	Technology	Purpose
Icons	Lucide React	Consistent, lightweight icon system

4.2 API Integrations

The platform integrates with multiple third-party APIs for real-time data. All API keys are configured via environment variables — no hardcoded keys are present in the source code.

Service	Data Type	Key Required	Fallback
CoinGecko	Crypto prices, market cap, No (free tier) sparklines		Demo data with realistic prices
Alpha Vantage	Stock quotes, symbol search	Yes (VITE_ALPHA_VANTAGE_KEY)	Mock stock data
Finnhub	Financial news	Yes (VITE_FINNHUB_KEY)	Curated demo news

API Key Setup: Create a .env file in the project root with your keys. The application gracefully degrades to demo data when keys are not provided or APIs are unavailable.

4.3 Data Persistence

All user data is stored in the browser's localStorage, enabling full functionality without a backend server. This architecture prioritizes simplicity and zero infrastructure costs while providing persistent user experience.

Storage Key	Data Type	Structure
verdexis_auth	User authentication state	{ email, name }
verdexis_holdings	Portfolio holdings	Array of holding objects with id, symbol, quantity, price, pnl
verdexis_wallet	Wallet balances	Array of balance objects with currency, symbol, balance, available
verdexis_trades	Trade history	Array of trade objects with id, symbol, side, price, quantity, timestamp
verdexis_transactions	Wallet transactions	Array of tx objects with id, type, amount, currency, description, status

Data flows: Trading execution updates both holdings and trades. Wallet operations update balances and transactions. All changes persist across page refreshes and browser sessions.

5. User Flows

5.1 New User Registration

- 0. Landing Page: User visits verdexis.com and views the hero section with value propositions
- 1. CTA Click: User clicks "Sign Up" button, opening the auth modal
- 2. Account Creation: User enters first name, last name, email, and password (min 8 chars)
- 3. Demo Data Seed: On successful registration, demo portfolio data is automatically created
- 4. Dashboard Redirect: User is taken to their personalized dashboard with \$388K demo portfolio
- 5. Onboarding: User can explore Dashboard, Trading, Wallet, and AI Analyst features

5.2 Trade Execution Flow

- 6. Market Selection: User navigates to Trading page and selects a cryptocurrency from the market list
- 7. Order Configuration: User selects buy/sell, order type (market/limit), and enters amount
- 8. Validation: System checks sufficient balance (authenticated users only)
- 9. Execution: Trade is recorded in localStorage, portfolio holdings are updated
- 10. Confirmation: Toast notification confirms the trade with details
- 11. Portfolio Update: Dashboard and Wallet reflect the new position automatically

5.3 Authentication-Gated Experience

The platform uses a tiered access model:

Access Level	Public (Anonymous)	Authenticated
Landing Page	Full access	Full access
Market Data	Real-time prices + sparklines	Full access
AI Insights	Viewable (limited)	Full access with portfolio context
Dashboard	Market overview only	Full portfolio + analytics
Trading	View markets, execute trades gated	Full trading with balance display
Wallet	Login prompt	Full deposit/withdraw/transfer
News	Full access	Full access

6. Security & Compliance

Verdexis implements multiple security layers to protect user data and ensure regulatory compliance.

6.1 Security Features

- 256-bit SSL Encryption: All data transmission is encrypted in transit
- Local Data Storage: No user data is transmitted to external servers — everything stays in the browser
- No API Key Exposure: All third-party API keys use environment variables with empty-string fallbacks
- Session Management: Auth state validated on every protected route access
- Input Validation: All form inputs validated client-side before processing

6.2 Compliance Standards

- SOC 2 Type II: System and Organization Controls certification
- ISO 27001: Information Security Management standard
- PCI DSS: Payment Card Industry Data Security Standard
- GDPR: General Data Protection Regulation compliance

Note: Compliance badges represent target certifications for the production platform. The current implementation is a frontend demonstration with localStorage-based data persistence.

7. Roadmap & Milestones

Phase	Timeline	Deliverables
Phase 1: MVP	Completed	Landing page, auth modal, dashboard, trading, wallet, AI insights, news
Phase 2: Backend	Q3 2026	Node.js/Express API, PostgreSQL database, JWT auth, real trade execution
Phase 3: Exchange APIs	Q4 2026	Live trading via Binance/Coinbase APIs, real order execution
Phase 4: Mobile	Q1 2027	React Native iOS/Android apps with feature parity
Phase 5: Advanced AI	Q2 2027	LLM integration (GPT-4), predictive

Phase	Timeline	Deliverables
		analytics, sentiment analysis
Phase 6: Social	Q3 2027	Copy trading, social feeds, leaderboards, community features

This specification document represents the current state of the Verdexis platform as of May 2026. All features described in Phase 1 are implemented and deployed. Subsequent phases represent planned enhancements for the production roadmap.

